

Simple and Compound Machines

Introduction: The Machines That Make Work Easier

People use machines every day without giving them a second thought. A pair of scissors cutting paper, a bicycle carrying a rider to school, a crane lifting heavy materials at a construction site — all of these are machines. Yet none of them have motors or engines in the traditional sense. They rely on fundamental principles of physics to multiply force, change direction, or increase speed.

Simple machines are basic mechanical devices that change the direction or magnitude of a force. There are six classical types: the lever, the wheel and axle, the pulley, the inclined plane, the wedge, and the screw. **Compound machines** are combinations of two or more simple machines working together to perform more complex tasks. Bicycles, scissors, wheelbarrows, and cranes are all examples of compound machines.

Understanding how these machines work requires knowledge of two fundamental types of motion: **translational motion** (straight-line movement) and **rotational motion** (circular movement around an axis). Most machines, especially compound ones, utilize both types of motion to accomplish their tasks efficiently.

This reading material examines the six types of simple machines, the distinction between simple and compound machines, the role of translational and rotational motion in machine function, and the forces (friction, weight, tension, and torque) that affect machine performance.

Part 1: Simple Machines

Definition of Simple Machines

Simple machines are basic mechanical devices that change the direction or magnitude of a force. They are the simplest mechanisms that use mechanical advantage to multiply force. No simple machine contains moving parts beyond its basic structure; they are the building blocks of all more complex machines.

The Six Types of Simple Machines

Type	Description	How It Works	Examples
Lever	A rigid bar that pivots around a fixed point called the fulcrum	Force applied at one point (effort) is multiplied to lift a load at another point	Seesaw, crowbar, scissors, pliers, bottle opener
Wheel and Axle	A larger wheel attached to a smaller axle, both rotating together	Force applied to the wheel rotates the axle, or force applied to the axle rotates the wheel	Doorknob, steering wheel, screwdriver, bicycle wheel
Pulley	A wheel with a grooved rim that holds a rope or cable	Changes the direction of force; multiple pulleys reduce the force needed to lift a load	Flagpole, window blinds, construction crane, elevator
Inclined Plane	A flat surface set at an angle to the horizontal	Allows a load to be raised gradually rather than lifted straight up, reducing required force	Ramp, staircase, sloping driveway, mountain road

Type	Description	How It Works	Examples
Wedge	A triangular-shaped tool with a thin edge	Splits or separates objects by converting force applied to its blunt end into force perpendicular to its sloping surfaces	Knife, axe, doorstop, chisel, nail
Screw	An inclined plane wrapped around a cylinder	Rotational motion is converted into linear motion; holds objects together or lifts materials	Jar lid, light bulb, vice, drill bit, bolt

How Simple Machines Provide Mechanical Advantage

Mechanical advantage is the factor by which a machine multiplies the force applied to it. A machine with a mechanical advantage greater than 1 reduces the amount of force needed to perform a task, even though the distance over which the force must be applied increases.

Simple Machine	Mechanical Advantage	Trade-Off
Lever	Depends on distances from fulcrum to effort and load	Longer effort arm = less force needed, but greater distance moved
Wheel and axle	Ratio of wheel radius to axle radius	Larger wheel = less force needed, but more rotations required
Pulley	Number of rope segments supporting the load	More pulleys = less force needed, but more rope must be pulled
Inclined plane	Ratio of slope length to height	Longer slope = less force needed, but longer distance traveled

Simple Machine	Mechanical Advantage	Trade-Off
Wedge	Ratio of wedge length to thickness	Longer, thinner wedge = less force needed, but greater insertion distance
Screw	Circumference of screw head divided by thread spacing	Finer threads = less force needed, but more rotations required

Part 2: Compound Machines

Definition of Compound Machines

Compound machines are devices that combine two or more simple machines to perform a task. By integrating multiple simple machines, compound machines can accomplish more complex functions than any single simple machine alone.

Examples of Compound Machines

Compound Machine	Simple Machines Involved	Function
Bicycle	Wheel and axle, lever, pulley (chain and gears), screw	Transfers rotational motion from pedals to wheels, propelling the bike forward
Scissors	Lever (two levers pivoting together) and wedge (blades)	Cutting force is multiplied through the lever action and concentrated on the wedge-shaped blades

Wheelbarrow	Lever (handles and wheel as fulcrum) and wheel and axle	Lifts and transports heavy loads with reduced effort
Crane	Pulley (cables), lever (boom), wheel and axle (base rotation), screw (stabilizers)	Lifts, rotates, and positions heavy loads precisely
Can opener	Lever (handle), wedge (cutting blade), wheel and axle (turning mechanism), screw (clamping)	Opens sealed cans by combining cutting, turning, and prying actions
Car	Wheel and axle (wheels), lever (steering wheel, brake pedals), screw (fasteners), pulley (engine belts)	Converts engine rotation into translational motion of the vehicle

Why Compound Machines Are More Efficient Than Simple Machines Alone

Reason	Explanation
Multiple functions	Compound machines can perform several tasks (lifting, moving, rotating) in a single device
Greater mechanical advantage	Combining simple machines multiplies the overall mechanical advantage
Directional flexibility	Motion can be redirected and transformed (rotational to translational, vertical to horizontal)
Force optimization	Different parts of the machine can apply force in different ways appropriate to each subtask
Precision and control	Compound machines allow finer control over speed, force, and position

Part 3: Translational and Rotational Motion in Machines

Translational Motion in Machines

Translational motion occurs when every point on an object moves the same distance in the same direction at the same time. In machines, translational motion is often the desired output — moving a load from one location to another.

Machine	Translational Component	Purpose
Bicycle	Forward movement of the entire bike	Transporting the rider from one place to another
Crane	Vertical lifting of the hook and load	Raising materials to higher elevations
Scissors	Closing motion of the blades	Cutting material placed between the blades
Elevator	Vertical movement of the cab	Transporting people or goods between floors

Rotational Motion in Machines

Rotational motion occurs when an object turns or spins around an axis. In machines, rotational motion often serves as the input — the motion that is transformed into useful work.

Machine	Rotational Component	Purpose
Bicycle	Wheels spinning, pedals rotating, chain moving around gears	Converting leg power into forward motion
Crane	Base rotation, pulley wheels spinning, cable drum turning	Positioning the load horizontally

Machine	Rotational Component	Purpose
Scissors	Pivot point rotation	Multiplying force applied to the handles
Wind turbine	Blades rotating around the hub	Capturing wind energy to generate electricity

How Machines Combine Both Motions

Most machines, especially compound ones, integrate both translational and rotational motion. The input motion (often rotational from a motor or human effort) is transformed into the desired output motion (often translational) through mechanisms such as gears, pulleys, levers, and screws.

Transformation	Mechanism	Example
Rotational to translational	Rack and pinion gear, screw	Steering wheel rotation turns car wheels; screw converts rotation into linear clamping force
Translational to rotational	Piston and crankshaft	Car engine pistons move linearly, rotating the crankshaft
Rotational to rotational (speed change)	Gears of different sizes	Bicycle gears allow the rider to climb hills with less effort
Rotational to rotational (direction change)	Bevel gears, pulleys	Power is transmitted around corners or to different locations

Part 4: Forces Affecting Machine Motion

Several forces influence how machines move and how efficiently they perform work.

Friction

Definition: A force that opposes motion between two surfaces in contact.

Effect on Machines

Reduces efficiency by converting useful work into heat

Causes wear and tear on moving parts

Can be helpful (brakes, tires gripping the road)

How to Manage

Lubrication (oil, grease) reduces friction

Use of ball bearings or rollers

Designing surfaces with appropriate texture

Examples in machines:

- Bicycle brakes rely on friction to slow the wheels
- Engine oil reduces friction between moving metal parts
- Tread on tires increases friction with the road for better traction

Weight (Gravity)

Definition: The force due to gravity acting on an object's mass.

Effect on Machines

Must be overcome to lift objects

Affects stability and balance

How to Manage

Use of mechanical advantage (levers, pulleys, inclined planes)

Designing wide bases and low centers of gravity

Influences acceleration and
stopping distance

Heavier loads require more force to move and more
distance to stop

Examples in machines:

- Cranes must be designed to support the weight of their loads
- Elevators use counterweights to balance the cab's weight
- Vehicles with heavy loads require more powerful brakes

Tension

Definition: A pulling force transmitted through a cable, rope, string, or similar flexible connector.

Effect on Machines

How to Manage

Transmits force over distances

Using strong, flexible materials (steel cables,
synthetic ropes)

Must be balanced to prevent breakage

Proper tensioning and regular inspection

Used to change direction of force

Pulleys redirect tension forces

Examples in machines:

- Crane cables are under high tension when lifting loads
- Bicycle brake cables transmit force from the lever to the brake pads
- Elevator cables support the weight of the cab and passengers

Torque

Definition: A rotational force that causes an object to rotate around an axis.

Effect on Machines	How to Manage
Causes rotational motion in wheels, gears, and axles	Applying force at a greater distance from the axis increases torque
Must be sufficient to overcome resistance	Using gears to multiply torque
Can cause twisting or shearing of components	Using materials with appropriate strength

Examples in machines:

- A longer wrench handle produces more torque for loosening tight bolts
- Bicycle pedals apply torque to the crank, which turns the chainring
- Car engines produce torque that is transmitted to the wheels

Part 5: Comparing Old and New Machine Designs

Technological advancements have significantly improved how machines use translational and rotational motion.

Machine	Old Design	Motion Limitations	Modern Design	Motion Improvements
Bicycle	Solid rubber tires, direct drive (pedals connected directly to wheel)	High rolling resistance, inefficient power transfer	Pneumatic tires, derailleur gears, chain drive	Reduced friction, optimized gear ratios for different terrain

Machine	Old Design	Motion Limitations	Modern Design	Motion Improvements
Wind turbine	Simple wooden blades, fixed pitch	Low efficiency, limited wind speed range	Aerodynamic composite blades, variable pitch control, yaw mechanism	Higher rotational efficiency, optimal blade angle at all wind speeds
	Manual or steam-powered, simple pulley systems	Slow operation, limited lifting capacity	Hydraulic systems, computer-controlled motors, telescoping booms	Precise control, high lifting capacity, combined rotational and translational motion
Car engine	Large, heavy, inefficient internal combustion	Poor fuel economy, high emissions	Turbocharging, hybrid systems, electric motors	Higher torque at lower RPM, regenerative braking (rotational to electrical energy)

How Modern Machines Convert Rotational Motion More Efficiently

Innovation	How It Works	Efficiency Gain
Variable transmission	Adjusts gear ratios continuously rather than in fixed steps	Engine operates at optimal RPM for a wider range of speeds
Regenerative braking	Converts rotational kinetic energy of wheels into electrical energy	Energy that would be lost as heat is captured and reused

Innovation	How It Works	Efficiency Gain
Aerodynamic blade design	Blade shape optimized using computer modeling	Wind turbines capture more energy from the same wind speed
Low-friction bearings	Precision bearings reduce rotational resistance	Less energy lost to heat, more energy delivered to output

Part 6: The Lever — A Closer Look

The lever is one of the simplest and most important simple machines. Understanding how it works provides insight into all machines that use rotational motion to multiply force.

Parts of a Lever

Part	Definition
Fulcrum	The fixed pivot point around which the lever rotates
Effort (input force)	The force applied to the lever
Load (output force)	The force exerted by the lever on the object being moved
Effort arm	Distance from the fulcrum to the point where effort is applied
Load arm	Distance from the fulcrum to the point where the load is applied

How a Lever Works

When force is applied to the effort arm, the lever rotates around the fulcrum. This rotational motion is then transmitted to the load arm, which applies force to the load. The mechanical advantage depends on the ratio of the effort arm length to the load arm length.

Fulcrum Position	Mechanical Advantage	Example
Closer to load	Greater than 1 (less force needed)	Wheelbarrow, nutcracker
Closer to effort	Less than 1 (more force needed, but greater speed or distance)	Fishing rod, human forearm
Exactly in the middle	Equal to 1 (force is neither multiplied nor reduced)	Seesaw, balance scale

Real-World Lever Applications

Application	Type of Lever	How Motion Is Used
Scissors	First-class lever (fulcrum between effort and load)	Rotational motion of handles is transferred to blades for cutting
Wheelbarrow	Second-class lever (load between fulcrum and effort)	Rotational motion of wheel allows lifting heavy loads with less force
Human arm (lifting a weight)	Third-class lever (effort between fulcrum and load)	Rotational motion at the elbow allows rapid movement of the hand

Part 7: The Wheel and Axle — A Closer Look

The wheel and axle is a simple machine consisting of a larger wheel attached to a smaller axle, both rotating together. Understanding how it works reveals how rotational motion can be used to multiply force or increase speed.

Parts of a Wheel and Axle

Part	Definition
Wheel	The larger circular component; force is often applied here
Axle	The smaller cylindrical component attached to the wheel's center; the axle rotates with the wheel
Effort (input force)	The force applied to either the wheel or the axle, depending on the application
Load (output force)	The force exerted by the opposite component (axle or wheel)
Radius of wheel	Distance from the center of rotation to the outer edge of the wheel
Radius of axle	Distance from the center of rotation to the outer surface of the axle

How a Wheel and Axle Works

When force is applied to the larger wheel, the axle rotates with the same angular velocity but with less linear force at its surface. Conversely, when force is applied to the smaller axle, the wheel rotates with greater linear force at its rim but with reduced angular velocity. The mechanical advantage depends on the ratio of the wheel radius to the axle radius.

Force Application	Mechanical Advantage	Example
Force applied to wheel	Greater than 1 (force is multiplied at the axle)	Screwdriver turning a screw (hand applies force to the handle, the axle turns the screw with greater force)
Force applied to axle	Less than 1 (speed is multiplied at the wheel)	Bicycle wheel (pedals apply force to the axle; the wheel rim moves faster than the axle)
Equal radii	Equal to 1 (no multiplication)	A solid disk with no distinct wheel and axle difference

Real-World Wheel and Axle Applications

Application	Force Application	How Motion Is Used
Doorknob	Force applied to wheel (knob)	Rotational motion of the knob turns the axle, which retracts the latch
Steering wheel	Force applied to wheel	Rotational motion of the large wheel turns the smaller steering column axle, allowing precise control of the vehicle's direction
Ferris wheel	Force applied to axle (by motor)	Rotational motion of the axle turns the large wheel, carrying passengers around the circle
Windlass (well crank)	Force applied to wheel (crank handle)	Rotational motion of the handle turns the axle, winding the rope and lifting the bucket
Gear train (two gears)	Force applied to larger gear	Rotational motion of the larger gear turns the smaller gear at higher speed but lower torque

Part 8: The Pulley — A Closer Look

The pulley is a simple machine consisting of a wheel with a grooved rim that holds a rope, cable, or belt. Understanding how it works reveals how tension forces can be redirected and multiplied to lift heavy loads.

Parts of a Pulley

Part	Definition
Wheel (sheave)	The grooved wheel around which the rope passes
Axle	The pin or shaft about which the wheel rotates
Rope (or cable)	The flexible connector that transmits tension force
Effort (input force)	The force applied to the rope (pulling end)
Load (output force)	The weight being lifted or moved
Support	The fixed structure holding the pulley axle in place

How a Pulley Works

When force is applied to the rope, the wheel rotates, reducing friction as the rope moves. The mechanical advantage depends on how many rope segments support the load. Each supporting rope segment carries an equal share of the load's weight.

Pulley Configuration	Mechanical Advantage	Example
Single fixed pulley	1 (force is neither multiplied nor reduced)	Flagpole; changes direction of force only

Pulley Configuration	Mechanical Advantage	Example
Single movable pulley	2 (load is supported by two rope segments)	Lifting a heavy object with a pulley attached to the load
Block and tackle (2 fixed, 2 movable)	4 (four rope segments support the load)	Construction crane, ship rigging
Block and tackle (multiple pulleys)	Number of rope segments supporting the load	Heavy lifting in garages and workshops

Types of Pulleys

Type	Description	Mechanical Advantage
Fixed pulley	Pulley axle is attached to a fixed support; changes direction of force only	$MA = 1$
Movable pulley	Pulley axle is attached to the load; multiplies force	$MA = 2$
Compound pulley (block and tackle)	Combination of fixed and movable pulleys	$MA = \text{number of rope segments supporting the load}$

Real-World Pulley Applications

Application	Pulley Type	How Motion Is Used
Flagpole	Fixed pulley	Rotational motion of the pulley changes the direction of the pulling force from downward to upward

Application	Pulley Type	How Motion Is Used
Construction crane	Compound pulley	Rotational motion of multiple pulleys distributes tension, allowing a small motor to lift extremely heavy loads
Window blinds	Fixed pulley (cord lock)	Rotational motion of the pulley changes direction; the lock mechanism holds the load
Elevator	Compound pulley (counterweight system)	Rotational motion of pulleys balances the cab's weight, reducing the force needed from the motor
Exercise cable machine	Fixed and movable pulleys	Rotational motion of pulleys redirects force, allowing exercises from different angles

Part 9: The Inclined Plane — A Closer Look

The inclined plane is a simple machine consisting of a flat surface set at an angle to the horizontal. Unlike the lever, wheel and axle, and pulley, the inclined plane does not involve rotational motion. It relies entirely on translational motion to reduce the force needed to raise an object.

Parts of an Inclined Plane

Part	Definition
Slope (ramp surface)	The flat, angled surface over which the object moves
Height (rise)	The vertical distance from the bottom to the top of the inclined plane
Length (run)	The horizontal distance along the base, or the distance along the slope

Part	Definition
Effort (input force)	The force applied parallel to the slope to move the object upward
Load (output force)	The weight of the object being moved (force due to gravity)
Angle of inclination	The angle between the slope and the horizontal surface

How an Inclined Plane Works

When an object is lifted straight up, the full weight of the object must be overcome. When the same object is moved up an inclined plane, the force required is reduced because the weight is distributed along the slope. The mechanical advantage depends on the ratio of the slope length to the height.

Slope Length vs. Height	Mechanical Advantage	Example
Slope much longer than height	Greater than 1 (less force needed, but longer distance)	Long, gentle ramp leading to a building entrance
Slope only slightly longer than height	Slightly greater than 1 (force only slightly reduced)	Steep driveway
Slope equal to height (vertical)	1 (no advantage; effectively lifting straight up)	Ladder (not an inclined plane)

Trade-Off of Using an Inclined Plane

Advantage	Disadvantage
Reduces the force needed to raise an object	Increases the distance over which the force must be applied

Safer than lifting straight up (less risk of dropping the load)

Takes more time to move the object the longer distance

Allows heavy objects to be moved without complex machinery

Requires space for the length of the slope

Real-World Inclined Plane Applications

Application	How Motion Is Used	Mechanical Benefit
Wheelchair ramp	Translational motion of the wheelchair up the slope	Reduces the force needed to raise the wheelchair to the entrance
Mountain road (switchbacks)	Translational motion of the vehicle along the winding road	Reduces the steepness of the climb, allowing vehicles to ascend gradually
Moving truck loading ramp	Translational motion of the hand truck or dolly up the slope	Allows heavy furniture to be loaded with less lifting force
Escalator	Translational motion of steps along the inclined track	Carries people upward with less effort than climbing stairs
Roller coaster lift hill	Translational motion of the train up the inclined track	Uses a chain to pull the train upward; gravity then provides the force for the descent

Part 10: The Wedge — A Closer Look

The wedge is a simple machine consisting of a triangular-shaped tool with a thin edge. It is essentially a moving inclined plane. Understanding how it works reveals how translational motion can be converted into splitting or separating forces.

Parts of a Wedge

Part	Definition
Blade (cutting edge)	The thin, sharp edge that first contacts the material
Back (blunt end)	The thick end where force is applied
Sloping surfaces	The two inclined planes that make up the wedge
Effort (input force)	The force applied to the blunt end (back) of the wedge
Output forces	The forces exerted perpendicular to the sloping surfaces, pushing the material apart

How a Wedge Works

When force is applied to the blunt end of the wedge, the wedge moves forward (translational motion). The sloping surfaces convert this forward force into forces directed perpendicular to the wedge faces, pushing the material apart. The mechanical advantage depends on the ratio of the wedge's length to its thickness.

Wedge Shape	Mechanical Advantage	Example
Long and thin (large length-to-thickness ratio)	Greater than 1 (large force multiplication)	Knife blade, razor, sewing needle

Wedge Shape	Mechanical Advantage	Example
Short and thick (small length-to-thickness ratio)	Slightly greater than 1 (moderate force multiplication)	Doorstop, chisel for rough work
Very thick (length nearly equal to thickness)	Approximately 1 (little force multiplication)	Wooden stake

Trade-Off of Using a Wedge

Advantage	Disadvantage
Multiplies the applied force significantly	Requires the wedge to be driven a long distance into the material
Can split materials that would be impossible to separate by hand	The wedge may become stuck if the material closes behind it
Simple and requires no moving parts	Once inserted, removing the wedge can be difficult

Types of Wedges

Type	Function	Examples
Cutting wedge	Separates material by shearing	Knife, scissors blade, axe, razor
Splitting wedge	Forces material apart along natural planes	Wood-splitting wedge, chisel
Holding wedge	Prevents movement or fills gaps	Doorstop, shim, nail
Piercing wedge	Penetrates material without splitting	Needle, nail, spear point

Real-World Wedge Applications

Application	How Motion Is Used	Mechanical Benefit
Knife cutting vegetables	Translational motion of the blade downward	The wedge shape concentrates force on a thin edge, cutting through the vegetable
Axe splitting wood	Translational motion of the axe head into the log	The wedge shape forces the wood fibers apart, splitting the log
Doorstop	Translational motion of the wedge under the door	The wedge shape converts horizontal force into upward force, lifting the door slightly and creating friction
Nail driven into wood	Translational motion of the nail (hammering)	The wedge-shaped point penetrates the wood fibers without splitting them excessively
Zipper teeth	Translational motion of the zipper slider	Wedge-shaped slider forces the interlocking teeth together or apart

Part 11: The Screw — A Closer Look

The screw is a simple machine consisting of an inclined plane wrapped around a cylinder. Understanding how it works reveals how rotational motion is converted into linear motion (translational motion) to hold objects together or lift materials.

Parts of a Screw

Part	Definition
Head	The top of the screw, where rotational force (torque) is applied

Part	Definition
Thread (helix)	The inclined plane wrapped around the cylinder; the spiral ridge
Pitch	The distance between adjacent threads (the vertical distance the screw advances in one full rotation)
Root	The solid central cylinder of the screw
Tip	The pointed end of the screw (if present)
Effort (input force)	The rotational force (torque) applied to the head of the screw
Load (output force)	The linear force (thrust) exerted by the screw as it advances into or through a material

How a Screw Works

When rotational force (torque) is applied to the screw head, the screw rotates. The inclined plane of the thread converts this rotational motion into translational motion, causing the screw to advance linearly into the material. The mechanical advantage depends on the circumference of the screw head divided by the pitch (distance between threads).

Thread Characteristic	Mechanical Advantage	Example
Fine threads (small pitch)	Greater than 1 (large force multiplication, slow advancement)	Machine screw, precision adjustment screw
Coarse threads (large pitch)	Moderate advantage (faster advancement, less force multiplication)	Wood screw, bolt
Very coarse threads (very large pitch)	Slightly greater than 1 (rapid advancement, minimal force)	Auger bit, drill bit

Thread Characteristic	Mechanical Advantage	Example
	multiplication)	

Trade-Off of Using a Screw

Advantage	Disadvantage
Provides very high mechanical advantage	Requires many rotations to advance a short distance
Can hold objects together with tremendous force	Threads can strip if over-tightened
Allows precise linear positioning (micrometer screws)	Requires a matching nut or threaded hole
Self-locking (will not back out easily without applied torque)	Can be difficult to remove if corroded

Types of Screws

Type	Function	Examples
Fastening screw	Holds two or more objects together	Wood screw, machine screw, bolt
Power transmission screw	Converts rotational motion into linear motion for moving loads	Lead screw (lathe), jack screw (car jack)
Adjustment screw	Precisely positions components	Micrometer screw, tuning peg on a guitar
Self-tapping screw	Cuts its own threads as it is driven	Sheet metal screw, drywall screw

Real-World Screw Applications

Application	How Motion Is Used	Mechanical Benefit
Jar lid	Rotational motion of the lid is converted into linear motion, sealing the lid against the jar rim	Allows a tight seal with moderate hand force
Light bulb	Rotational motion of the bulb is converted into linear motion, engaging the threads in the socket	Secures the bulb while allowing easy removal
Car jack (scissor jack)	Rotational motion of the screw (turning the handle) is converted into linear motion, lifting the vehicle	A small rotational force lifts the heavy weight of the car
Vice (clamp)	Rotational motion of the screw handle is converted into linear motion, closing the jaws	Holds objects firmly with tremendous force
Drill bit (auger)	Rotational motion of the bit is converted into linear motion as the bit advances into the material	Cuts through wood or metal with relatively little downward force
Guitar tuning peg	Rotational motion of the peg is converted into linear motion of the string (changing tension)	Allows precise adjustment of string pitch

Part 12: Forces in Simple Machines

Each simple machine experiences characteristic forces that affect its performance.

Simple Machine	Primary Forces	How Forces Affect Motion
Lever	Torque (rotational force), weight of load	Torque is multiplied based on lever arm lengths; heavier loads require greater effort or longer effort arm
Wheel and axle	Torque, friction (at axle), weight	Larger wheel diameter reduces effort force; friction at axle reduces efficiency
Pulley	Tension (in rope), weight of load, friction	Multiple pulleys distribute tension, reducing required force; rope friction affects smooth operation
Inclined plane	Weight (component parallel to slope), friction	Longer slope reduces required force but increases distance; friction opposes motion up the slope
Wedge	Force applied to blunt end, friction on sloping surfaces	Force is converted perpendicular to the wedge faces; friction determines how easily the wedge penetrates
Screw	Torque (applied to head), friction on threads, tension	Rotational motion is converted into linear motion; finer threads provide greater mechanical advantage

Key Terms

Term	Definition
Simple machine	A basic mechanical device that changes the direction or magnitude of a force
Compound machine	A device that combines two or more simple machines
Mechanical advantage	The factor by which a machine multiplies the force applied to it
Lever	A rigid bar that pivots around a fixed point called the fulcrum
Wheel and axle	A larger wheel attached to a smaller axle, both rotating together
Pulley	A wheel with a grooved rim that holds a rope or cable
Inclined plane	A flat surface set at an angle to the horizontal
Wedge	A triangular-shaped tool with a thin edge
Screw	An inclined plane wrapped around a cylinder
Friction	A force that opposes motion between two surfaces in contact
Tension	A pulling force transmitted through a cable, rope, or similar flexible connector
Torque	A rotational force that causes an object to rotate around an axis
Translational motion	Motion in which every point on an object moves the same distance in the same direction at the same time
Rotational motion	Motion in which an object turns or spins around an axis

Review Questions

Responses should be written on a separate sheet of paper.

1. List the six types of simple machines and provide one example of each.
2. What is the difference between a **simple machine** and a **compound machine**? Provide one example of each.
3. A bicycle uses both translational and rotational motion. Identify which parts of a bicycle exhibit rotational motion and which exhibit translational motion.
4. Explain how a **lever** provides mechanical advantage. How does the position of the fulcrum affect the force needed to lift a load?
5. A crane uses a pulley system to lift heavy loads. What force is primarily at work in the cables? How does adding more pulleys affect the force needed to lift the load?